Optimization of Space Shuttle External Tank Design Return On Investment (ROI)

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Abstract

Our project aims and seeks to maximize the Return on Investment (ROI) for the space shuttle by using the optimization techniques on the External Fuel Tank (EFT) design. This method throws light on adjusting the design variablessuch as weight, aerodynamic features, and materials- while ensuring that the tank meets its important and useful mechanical and physical demands, including its volume, structural stress, and vibration stability. Optimization techniques like the Sequential Least Square Quadratic Programming (SLSQP) are used to obtain the best design options available. By reducing the tank's weight and cost without compromising its structural integrity, the project aims to improve the overall profitability and payload capacity of each space mission organized by the country.

Keywords: Maximize Return On Investment, External Fuel Tank, Sequential Least Square Quadratic Programming,Genetic Algorithm(GA), Saab's Lightweight Torpedo system, Machine Learning, structural stability, vibration, stress, vibration factor.

1 Introduction

The external space shuttle tank plays an important role in the mission and improving its design is of importance in performance and cost. Initial studies by Harold Coldwater[1] and Michael P. Nemeth [2] identified the possibility of reducing their weights, and the significance of the nonlinear analysis on fairness of the structures. The work of Coldwater demonstrated that Optimization of the design could save 7,294 pounds of weight, a major performance improvement. Nemeth stressed that nonlinear structural responses can be hazardous and that only small imperfections are needed.

Expanding on these pioneer works and to a great extent operating on the traditional optimization, Hsieh and Lawson[3] showed that both performance and cost optimization could be attained by adopting methods such as GA and multi-objective genetic algorithms. Our project takes this study further by applying the SLSQP technique specially on maximizing the investment on space missions ROI. The main aim is to determine the optimal parameters of the design stage of the external tank to ensure a higher ROI and structural integrity as well as stability even in the presence of maximum stress and dynamic responses. We integrate through combination of analysis of different tank designs with machine learning to make profits predictions filling the gap between engineering perfection and economics. The key goals of this project include the following

1.1 OPTIMIZING THE TANK DESIGN:

We consider the values of the variables and the parameters such as the values of the height, radius, length, and wall thickness as showed in Fig 1, to calculate the configurations that balance the fuel efficiency, materials used, and the structural integrity of the space shuttle fuel tank."

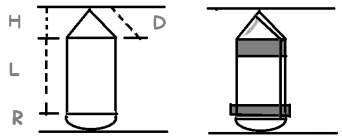


Fig. 1 The design variables used in the calculations.

The above Image is a small depiction of the design variables like the height, length, depth and the radius used for the Space Shuttle External Fuel Tank which are the key elements in the calculation of the efficiency and ROI.

1.2 COST AND STRESS CONSTRAINTS

The design of the tank must meet some certain stress and the vibrational limits in order to ensure the structural safety of the shuttle in the launch conditions.

The above Image (left) shows the intertank of the space shuttle which will be placed inside the External fuel tank satisfying the cost and stress constraint. The right



Fig. 2 Intertank structure. Adapted from [2]

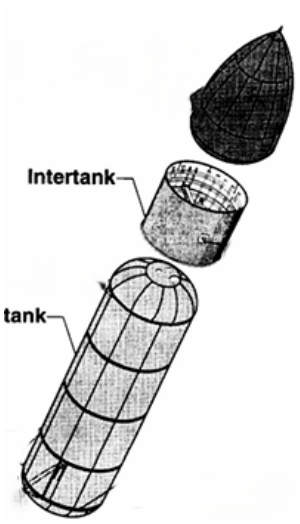


Fig. 3 External fuel tank with main and inter tank. Adapted from [2]

image shows the external fuel tank with the main tank and the inter tank in Fig. 2, on which the ROI is calculated based on the design variables in Fig. 1.

1.3 RETURN ON INVESTMENT

The ROI is the main goal of this project, and given the Trade Offs and the Tradeoffs related to the payload capacity, the launch costs, and the material expenses will have to be minimized. To do so, we make use of machine learning, with a regression model trained by synthetic data able to predict ROI given different tank configurations. This predictive tool allows the efficient validation of design and iterative optimization, which foresees the result without re-calculations. The combination of machine learning and natural means of main optimization methods such as SLSQP will become a data-driven methodology of cost- and sustainable aerospace design. The study demonstrates that state of the art computation techniques can be used to facilitate improved decision-making on complicated engineering tasks. The brief idea on usage of SLSQP in the project is given ahead: In the project this method is used to optimize the design of the external tank.

2 OBJECTIVE FUNCTION AND CONSTRAINTS

The objective function in the project aims to Maximize the ROI for the space shuttle tank design. Here the ROI is a measurement and a criterion to balance the payload capacity against all the costs of manufacturing and launching of the space shuttles which can be expressed mathematically in terms of the payload and the total cost

since ROI is essentially a measure of the payload efficiency per unit cost.

$$ROI = \frac{Payload\ Delivered}{Total\ cost} \qquad ROI = \frac{\pi P_{\text{launched}}}{C_{\text{tank}} + C_{\text{launch}}} \tag{1}$$

where: pi = Constant multiplier (scale the ROI formula); P_launched = payload of the oxygen + payload of the hydrogen(kg); C_tan = Total cost of the tank (Rs); C_launch = fixed launching cost per unit weight (Rs)
This is calculated on the basis of various factors like the payload weight, material and launch costs. In this project the function is designed to give the minimum of negative ROI in order to maximize the positive ROI since the optimization methods generally minimize the objective function. i.e. Min(−ROI). The tanks complex design must satisfy many of the important performance and engineering constrains to ensure the efficiency and the feasibility for which we need the constraint functions to keep a check on them,the used constraint functions are:

2.1 Volume constraints:

$$V_{\text{total}} = V_{\text{nose}} + V_{\text{cyl}} + V_{\text{hemisphere}} \geq V_{\text{nom}} \tag{2}$$

where:-

$$V_{\text{nose}} = \tfrac{1}{3}\pi R^2 H \quad \text{(cone)}, \quad V_{\text{cyl}} = \pi R^2 H \quad \text{(cylinder)}, \quad V_{\text{hemisphere}} = \tfrac{2}{3}\pi R^3 \quad \text{(hemisphere)} \tag{3}$$

Here the total internal volume of the tank must and should accommodate the required payload of the fuel because of which the formula has come into use.

2.2 Stress constraints:

$$\sigma_{\text{cylinder}} = \frac{P_{\text{fuel}} R}{2 t_{\text{cyl}}} \tag{4}$$

where:-

$$\sigma_{\text{cylinder}} \leq \sigma_{\text{allowed}} \tag{5}$$

P_fuel = Internal Pressure; t_cyl = Thickness of the cylindrical shell

Here the fuel tank should be able to withstand the internal pressure that is produced by the fuel without exceeding the allowable stress of materials of the tank.

2.3 Vibrational constraints:

$$\text{VibrationalFactor} = \frac{1}{H\,R\,L} \geq MinVibFactor \tag{6}$$

where:-
H = height of the nose cone(m); L = length of the cylindrical body (m); R = radius of the hemisphere (m); MinVibFactor = predefined value with respect to the dynamics of the launch vehicle

3 METHODOLOGY

We have been using a two-method approach strategy that uses SLSQP and Machine Learning (ML) to optimize external fuel tank design while optimizing the ROI. This approach makes the use of advantages of predictive analytics for assessing economic viability and gradient-based optimization for accurate design refinement. SLSQP as an optimization technique that is chosen as it can efficiently manage nonlinear constraints while minimizing a clearly stated objective function. By considering variables like payload capacity, material costs, and manufacturing charges, the aim function in this project was created to maximize return on investment. Among the design factors we have -the tank's height, radius, and length, which were geometric parameters. The conical, cylindrical, and hemispherical section's thicknesses are the examples of material properties.

3.0.1 Limitations:-

In order to comply with aeronautics specifications, the optimization process included major constraints: the volume constraint that provided enough fuel capacity (nominally 1200 m^3), structural constraints that restricted stress to withstand heat loads, aerodynamic stress, and launch-time vibrations. A vibration factor constraint was also required to keep structural integrity at a minimum level above. At each stage of the SLSQP algorithm, such constraints were imposed while checking the objective function, with design variables manipulated until an optimal compromise between performance, cost, and weight was met within all given limits.

3.1 Machine Learning Integration

Machine learning has been used to forecast the ROI for several design configurations in order to support the optimization process. By reducing the computing effort and providing insights into objective trade-offs, thus predicting capability expedited the examination of various possibilities.

3.1.1 Generation of Data:

To mimic the performance of several tank designs, synthetic data has been generated. Values for the material and geometric variables have been provided in every design configuration, the ROI was then determined using the objective function.

3.1.2 Training the model:

This artificial dataset that was generated has been used to build a regression model that predicted the ROI for some novel design configurations. To guarantee reliable predictions, the model was assessed using some metrics of which include The Mean Absolute Error, Mean Squared Error and R-squared (R^2).

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - y_i'| \qquad (7) \qquad\qquad \text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - y_i')^2 \qquad (8)$$

where:-
y_i = Actual value of the ith data point; y_i' = Predicted value of the ith data point; N = Number of data points

Benefits of ML Integration: Design configurations have been evaluated more quickly which has helped us to get a better comprehension of how design factors have affected the return on investment. This framework has allowed high ROI design exploration without the need for any manual-recalculation.

3.2 Code Flow and the Functionality

3.2.1 Objective Function:

the most important parameter to be optimized is ROI with payload, manufacturing and material costs consideration for the improved output and functionality.

3.2.2 Constraints

In this case, critical constraints are volume, stress and vibration parameters which must be specified for the design to meet the safety criteria also with regard to performance. These need to be learned every time while the optimization function is run through in a cycle.

3.2.3 Optimization Algorithm SLSQP:

The SLSQP technique iterates and improves the estimation of initial design variables, x0 by minimizing the negative ROI using the function "SciPy.Optimize.minimize". The approach algorithm checks the objective function on every step and is guaranteed that at that point all constraints are satisfied, the objective function is gradually converging toward an Optimal solution.

3.2.4 Machine Learning Model:

Synthetic data was generated analyzing the objective function for thousands of design configurations. Using that synthetic data, a regression model (perhaps linear regression) is built to predict the ROI for new designs. Measures of evaluation use predictions against actuals, and plots are generated containing scatter plot of predicted against real ROI.

3.2.5 Visualization and Insights:

Graphs that are used to explain the results in a better way which include the best tank design and the accuracy of the ML model and giving optimal.

3.3 Computing Tools

The optimization and machine learning process was conducted through the application of Python with a number of prominent libraries. SciPy was utilized for the development of the SLSQP optimization algorithm, whereas Scikit-learn served for training and evaluation of the machine learning model. Matplotlib and NumPy also facilitated

numerical computation and visualization. This approach thus gives a complete solution to the optimal design of the external fuel tank by integrating the predictive power of machine learning with the accuracy of SLSQP for constrained optimization. It structures the flow to ensure transparency and efficiency and opens room for further improvement.

3.4 Straties for Simulatation

To make the simulations more effective, advanced tactics were used in combination with optimization and machine learning. Synthetic data confirmed robustness under extreme conditions, and dynamic constraint adjustments highlighted aspects such as stress and vibration. Noise was introduced to material and cost parameters to simulate variability in reality, and batched runs enhanced computational efficiency. Sensitivity analysis determined critical design parameters impacting ROI, and scenario-based simulations with different fuels and payloads simulated adaptability. Lastly, the SLSQP algorithm was stress-tested with rigorous extreme constraints to ensure stability, scalability, and real-world applicability of the framework.

4 SIMULATION OUTCOMES

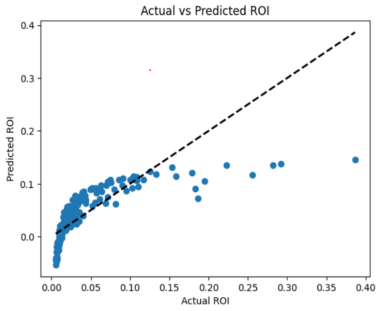


Fig. 4 Actually vs Predicted

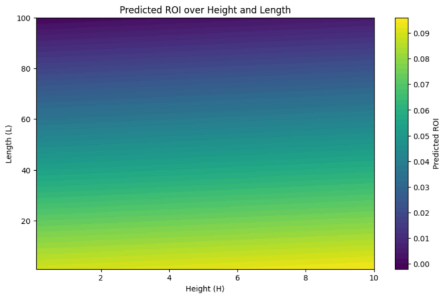


Fig. 5 contour plot

Above are the Results depicting the values of the regression coefficients and other important variables shows us the different types of errors along with the model accuracy.

The picture below shows the contour plot of the length and the height and how it affects the ROI, which is visible with the bar given on the right side.

MAE: 0.028826 MSE: 0.002240 R^2: 0.546316 Accuracy: 54.631421%

The values of variables taken for the different design variables of the space shuttle external fuel tank are given below: Optimal design variables (H, L, R, t, t_cy, t_co):

$H : 10.0; L : 2.5; R : 0.2; t : 0.001; t_cy : 0.001; t_co : 0.001; OptimalROI : 0.4434690433561773$

5 CONCLUSION & FUTURE SCOPE

Our project offers an insightful method of optimizing the Space Shuttle EFT through a combination of engineering design and data-driven decision-making, utilizing SLSQP and ML to achieve technical performance and economic feasibility. The main emphasis is to optimize ROI in space travels by optimizing key design parameters like weight, materials, and structural stability to achieve stress limits, vibration stability, and fuel capacity with reduced manufacturing expense without any compromise on safety and functionality. In the future, the architecture can be refined by utilizing high-performance materials such as ultra-lightweight alloys and nanostructures to enhance structural and thermal efficiency further and by using multi-objective and dynamic optimization techniques to resolve intricate nonlinear relationships and facilitate adaptive design in manufacturing. Moreover, incorporating sustainability measures and applying the framework to other parts of aerospace through additive manufacturing and stochastic optimization can provide both economic viability and environmental adaptability, thus developing the scope of cost-effective and sustainable aerospace engineering.

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